

Introduction to B-Tree & B+ Tree Data Structures

> **Definition:** A **B-Tree** is a self-balancing multi-way search tree of order m where each node can contain multiple keys and more than two children, designed specifically to optimize **disk read/write block accesses** in database indexing and file systems.

1. Detailed Technical Explanation

1. Properties of a B-Tree of Order m :

1. Every node has at most m children.
2. Every other node, except the root and leaf nodes, must be full. That is, it must contain m pointers to its children.

2. B-Tree vs B+ Tree Comparison

B+ TREE:

Detailed Comparison Table:

Feature	B-Tree	B+ Tree
Search Path	Search path varies depending on the key being searched for.	Search path is the same for all keys, as all keys are stored in the leaf nodes.

3. Core Concepts & Memory Keywords

- **Order m:** Maximum number of children a node can have.

- **F**

4. Must-Write Points for Exams

- B-Trees and B+ Trees reduce disk I/O by fitting large numbers of keys inside a single disk page block (e.g., 4KB or 8KB).

- **B+**

5. Quick Recall Flow

Multi-V